

# **Safe Route Xplorer**

Project Team

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# Chapter 1

## Introduction

The purpose of this document is to provide guidelines for a **Real-Time Road Safety Alert System** that will improve traveler and tourist safety. The application provides early alerts about potential traffic risks resulting from various disasters by integrating user-generated data from news channels and social media sites. Its purpose is to provide users with up-to-date information on the safety index of their routes so they may make wise judgements and stay out of harm's way.

The intended audience for this document includes various stakeholders:

- **Developers** will utilize this document to understand the technical requirements and functionalities needed for the application.
- **Project Managers** will refer to this document to ensure that the project stays within its defined scope and meets its objectives.
- **Marketing Staff** will use the information to develop promotional materials that accurately represent the application's features and benefits.
- **Users** will benefit from an overview of how the system operates and how it can enhance their travel experiences.
- **Testers** will refer to the requirements to develop test cases ensuring that the system functions as intended.

### 1.1 Problem Statement

Unexpected transportation dangers brought on by landslides, floods, and earthquakes commonly affect travellers. The inability of traditional guidance systems to promptly

notify users of these hazards can lead to dangerous circumstances, traffic jams, and accidents. Typical GPS systems do not account for unexpected changes in road safety, therefore the absence of real-time information might result in harmful scenarios. Another area where existing systems fail is the potential for user-generated data from social media sites, which can provide relevant and timely information on local conditions. As a result, guests and visitors face unnecessary risks since they are unable to obtain reliable information to assess the safety of their travels.

This research aims to bridge that gap by developing a \textbf{Real-Time Road Safety Alert System} that uses natural language processing (NLP) to filter social media postings for relevant dangers and update the safety index of different routes in real-time. By providing these functions, the system seeks to increase user comfort and safety while commuting by enabling users to take preventative action and develop well-informed opinions.

## 1.2 Scope

The Real-Time Road Safety Alert System offers current information on road hazards resulting from natural catastrophes to improve travel safety. The main job of the system is to look for user-generated material about road conditions on different news sites and social media platforms. The system will employ natural language processing (NLP) to look for terms in postings that could be danger indicators. This will enable the system to assess these risks using a predetermined range that spans from safe to critical. Because the safety indices for different routes will be updated in real-time, users will always have access to the most recent information on their travel routes.

Users will be able to view their routes and the associated safety index clearly using a map-based interface. The technology will provide users with alerts when major threats are detected, in addition to real-time updates, enabling them to plan their travel routes with the necessary information. The program will use Google Maps for areas with dependable internet access; offline users will get the most recent updates that are stored on their devices. The program will provide users with the last safe paths in the event that internet connectivity is not accessible, even if its main focus will be on external data sources.

Another part of the project's scope is the integration of an intuitive interface that allows passengers to customise their alerts and notifications to their preferences. The project scope does not presently include full danger checking, but the system will broadcast high-severity alarms to assure user safety. The objective is to offer a robust and workable solution that improves passenger experience while lowering the risks associated with unforeseen road hazards. This comprehensive plan will let clients travel with confidence and steer clear of dangerous situations.

## 1.3 Modules

### 1.3.1 Client Mobile App

The Client Mobile App allows access to information about safety routes, provides timely alerts, and even retrieves important information despite the fact that it is offline.

**1. User Interface (UI):**

- (a) Mobile-friendly map with route safety information
- (b) Setting: Zooming and swiping inwards and outwards on the mobile to make easy navigation through it
- (c) An intuitive and simple UI that will help navigate to specific information or any road condition
- (d) A search bar to help locate specific routes or locations.

**2. Route Safety Index:**

- (a) Mobile-friendly safety index display for chosen routes.
- (b) Dynamic updates of route safety on the mobile interface.
- (c) Simplified color-coded route indicators for quick decision-making.

**3. Real-time Alerts:**

- (a) Push notifications that are delivered directly to the user's mobile device.
- (b) Customized alert settings friendly for usage on the mobile phone.
- (c) Banners or pop-ups are displayed in the application as alerts.

**4. Offline Ability:**

- (a) Caching of information of routes already viewed accessible to users who are disconnected from the internet
- (b) There are alerts which can be stored for viewing at later times when the app is offline.

### **1.3.2 Client Web App**

The Client Web App would enable users to see an interactive interface as well as an informative interface regarding the safety of the route and real-time alerts.

#### **1. User Interface (UI):**

- (a) Interactive map to show information about the route safety.
- (b) Visual indications for identifying safe as well as unsafe routes.
- (c) Search option to find and get hold of exact routes or places
- (d) Real-time road condition updates for display.
- (e) A separate notification panel for alerts and warnings relating to safety

#### **2. Route Safety Index:**

- (a) The real-time data can be used to provide a safety index of routes selected.
- (b) Change the safety index in real-time with incoming data.
- (c) Color-coded routes for indication of different safety levels.

#### **3. Real-time Alerts:**

- (a) Push notifications with road hazards or safety alerts
- (b) Options to personalize the alerts preferences on customizations.
- (c) Alerts are shown as pop up on the user interface.

### 1.3.3 Admin Web App

Admin Web App: A central point that manages the data in our system and how things are performing, with overseeing user interaction. This is where we would have system maintenance and content moderation tools

#### 1. Data Management:

- (a) Checking on the incoming social media and news data through the admin interface
- (b) Tools for manually adjusting route safety indexes if needed
- (c) Historical data trends and user activity on the Dashboard

#### 2. System Monitoring:

- (a) Benchmarking and data processing tracker of the real-time system performance.
- (b) System malfunction error messages, e.g. data inconsistency or API call failure.
- (c) System activity audit-all actions recorded.

#### 3. User Management:

- (a) Account management for users with roles and permissions
- (b) User statistics and view of system activity
- (c) Tools that may be utilized to address some of the needs or concerns of the users-grandparents

#### 4. Content Moderation:

- (a) Filtering and regulation tool of the content fetched from other social media.
- (b) Highlight or delete feature of irrelevant or inappropriate content
- (c) Keywords as well as with sentiment analysis can enhance relevance of the content.

## 1.4 User Classes and Characteristics

This table structure clearly defines every user class, centered about their relationship. Individually with the system and focusing on how they interact with each core functionality. This will help developers and testers understand what users need and what users are expecting. It is designed and tested in these two phases.

| User Class                 | Description                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Traveler<br>(Regular User) | Travelers or tourists availing themselves of the system with the exact and real-time road safety update. They use GPS-based navigation to ensure safety while traveling and fetch Warnings on existence of dangers such as landslides, floods or road blockages. They may Access the system through mobile or web applications and are interested in Updated traffic and safety index data. |
| Commuter                   | Daily commuters for whom travel distance is not necessarily long, but for whom the system would be crucial for avoiding hazards or traffic problems during normal travels, say going to work. The service would mainly be used during peak hours, hence would be regularly updated.                                                                                                         |
| Road Safety Authorities    | Authorities with road safety portfolios would make use of the system to monitor the frequency and number of hazards and pinpointed areas. They will also apply it to send assistance or inform the public of progress made in the recovery process. Users need more comprehensive reports and analysis grounded on real-time and historical data.                                           |
| System Administrator       | These are system administrators ensuring the backend proper working, server health, and smooth processing of data, including API keys, user data, and logs. They manage inflow and outflow of data from external sources, like social media. They oversee system uptime and do a fair share of troubleshooting and maintenance activities.                                                  |
| Data Analyst               | Such analysts make use of the system to collect insights about how hazards affect people's travel behavior, analyzing both the volume and nature of the hazards at different locations. Analysts may even analyze patterns to enhance system algorithms and road safety practices. They work closely with NLP data, such as keyword analysis, for evaluating system efficiency.             |

## 2. Project Requirements

This chapter describes the functional and non-functional requirements of the project.

### 2.1 Functional Requirements

#### 2.1.1 Feature 1: Safe Route Planning

**Description:** The first feature of SafeRouteExplorer will allow users to enter their destination and receive the safest route based on real-time social media data.

- (a) The system shall allow the user to enter their starting location and destination.
- (b) The system shall show the safest route based on the data received and processing done on it.
- (c) The system shall dynamically update the map if a road hazard occurs along the route.
- (d) The system shall display the estimated time to reach the destination.

#### 2.1.2 Feature 2: Real-time Social Media Monitoring for Potential Hazards

**Description:** This feature will monitor social media and news sources for potential road hazards like accidents, crimes, natural disasters, etc.

- (a) The system shall continuously fetch tweets, Facebook, and Instagram posts in real-time based on specific keywords.
- (b) The system shall process and filter the incoming data to extract relevant information.
- (c) The system shall identify the severity of the hazard detected.
- (d) The system shall identify the location where the potential road hazard occurred.
- (e) The system shall classify the severity of the hazard and update the map to show the safest route for the user's destination.

### **2.1.3 Feature 3: Map Updating and Display**

**Description:** This feature will update the map and display the safest route after processing road hazard data.

- (a) The system shall update the routes if new hazards are dynamically detected.
- (b) The system shall remove or not display hazardous routes as the top route.

### **2.1.4 Feature 4: User Notifications**

**Description:** This feature allows real-time notifications to be sent to users if any hazards are detected and subsequently updates the maps.

- (a) The system shall send notifications to the user whenever their route is updated.
- (b) The system shall send alerts when hazards are detected along the planned route.

### **2.1.5 Feature 5: System Administration**

**Description:** This feature enables system administrators to manually manage maps and remove paths if necessary.

- (a) The system shall allow the administrator to manually remove paths if they are potentially dangerous.
- (b) The system shall allow the administrator to set severity levels and update the map.

## **2.2 Non-Functional Requirements**

### **2.2.1 Reliability**

The system should be highly reliable, ensuring the safest route is always shown and any failure is quickly handled.

### **2.2.2 Usability**

The system should be user-friendly, providing an interactive, visually appealing, and easy-to-navigate map for tourists.

### **2.2.3 Performance**

The system shall ensure smooth functionality, especially when a hazardous activity occurs or during high data traffic. It should respond smoothly even during peak hours.

### **2.2.4 Security**

The system shall maintain integrity by ensuring no unauthorized access to routes or data is allowed.

### **2.2.5 Availability**

The system should be available at all times to detect any hazardous activity on roads and update routes accordingly.

## **2.3 Domain Model**

The domain model for the Real-Time Safe Path Finder project is provided in below page

## **2.4 Use-case/Event Response Table/Storyboarding**

### **2.4.1 Use Case Diagrams**

The use case diagram for the system is provided below. This diagram illustrates the key interactions between users and the system, along with the major processes.

### **2.4.2 Detailed Use Cases**

**Use Case #1: Enter Destination**

**Use Case #2: Monitor Social Media and News for Road Hazards**

**Use Case #3: Update Maps Based on Data**

### **2.4.3 Event-Response Table**

The Event-Response table is provided below

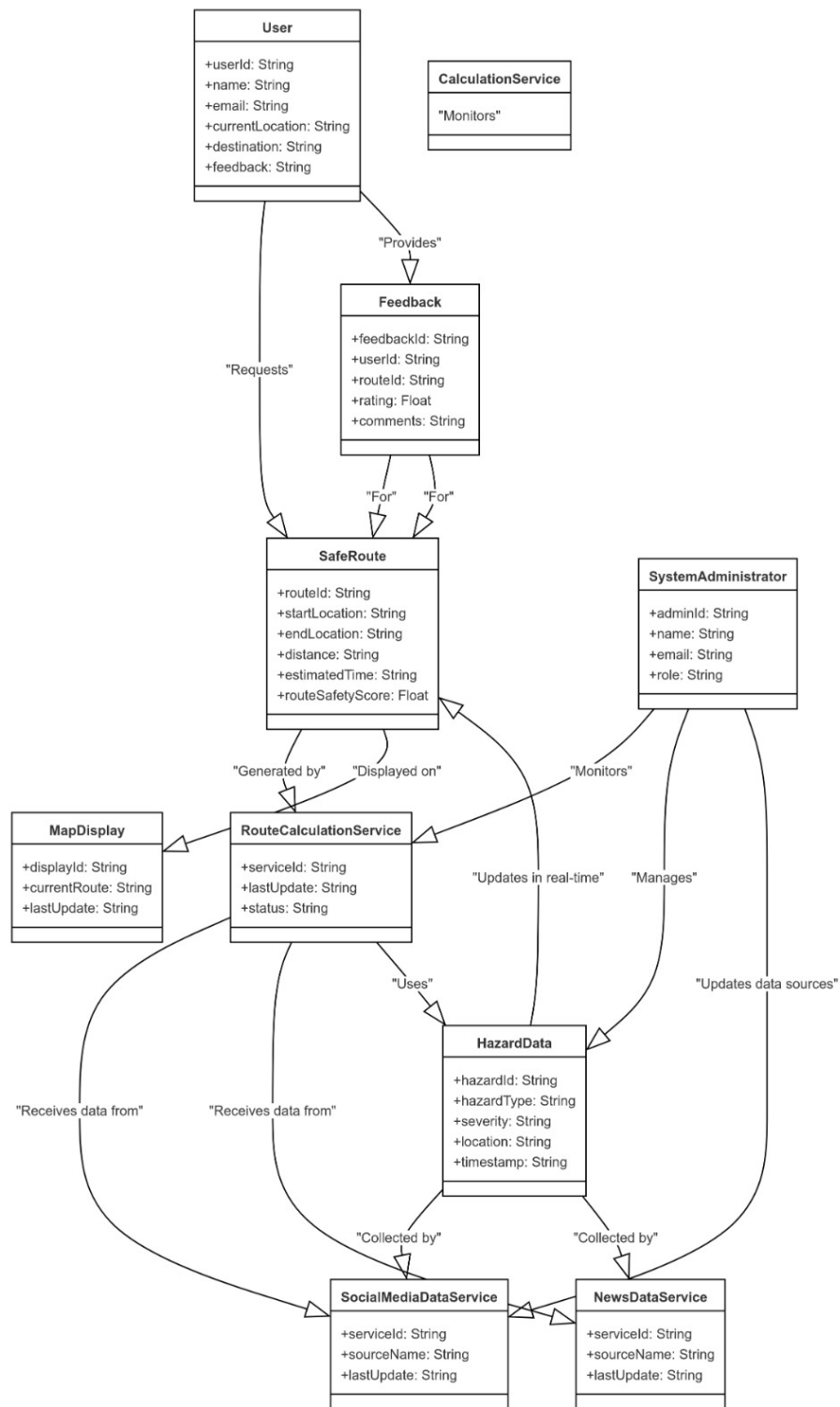


Figure 2.1: Domain Model for Real-Time Safe Path Finder

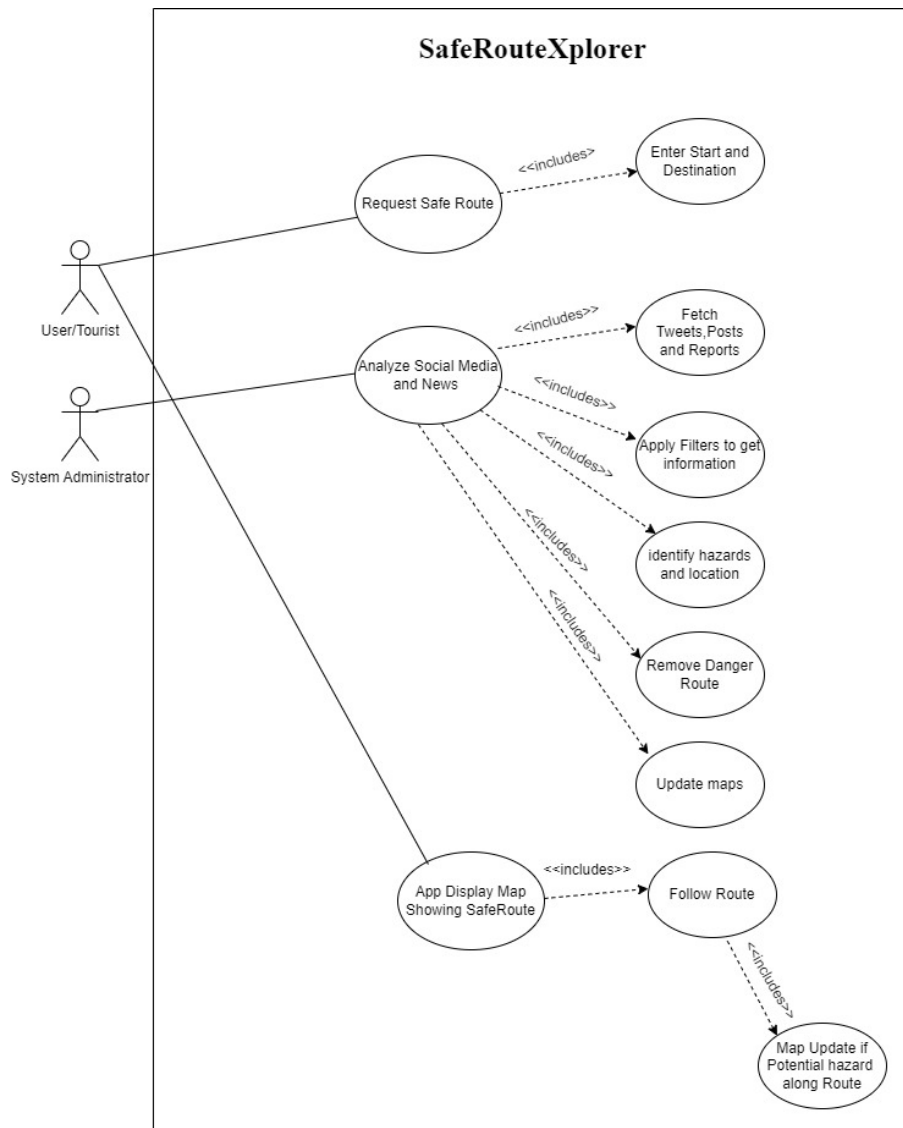


Figure 2.2: Use Case Diagram for Real-Time Safe Path Finder

|                     |                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors:</b>      | User                                                                                                                                                   |
| <b>Type:</b>        | Primary                                                                                                                                                |
| <b>Description:</b> | The User will enter their destination on the app, and the app will show the safest route, excluding any unsafe route based on social media monitoring. |

Table 2.1: Use Case #1: Enter Destination

|                     |                                                                                                                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors:</b>      | System                                                                                                                                                                                              |
| <b>Type:</b>        | Primary                                                                                                                                                                                             |
| <b>Description:</b> | The system will search for any road hazards on social media and news sites based on specific keywords. It will check the location and the severity of the issue to help travelers avoid any danger. |

Table 2.2: Use Case #2: Monitor Social Media and News for Road Hazards

|                     |                                                                                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors:</b>      | System                                                                                                                                                          |
| <b>Type:</b>        | Primary                                                                                                                                                         |
| <b>Description:</b> | The system will update maps based on the data it fetched after processing. It will omit routes that are potentially dangerous based on posts, tweets, and news. |

Table 2.3: Use Case #3: Update Maps Based on Data

| <b>Event</b>                                                | <b>Trigger</b>                                                       | <b>Response</b>                                                                                     | <b>System Component Responsible</b>                         |
|-------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| The user enters the start and destination                   | The location is shown on the map                                     | The system fetches the initial route for the destination                                            | Route Calculation                                           |
| The system monitors social media for potential road hazards | Loop running in the background for potential posts based on keywords | The system processes and filters the incoming stream of data, identifying the location and severity | Social Media Data Collection API                            |
| Hazard detection in the news                                | The system finds news related to road hazards based on keywords      | The system evaluates the severity and location of the news detected                                 | News Scraper                                                |
| The system updates the map with a safer route               | Potential hazards detected on the path                               | The system's algorithm assigns a safety index to paths and updates the paths with the safest route  | Algorithm to update routes                                  |
| Hazard no longer persistent in the area                     | The system detects the route is safe again for travel                | The system updates the map, removing hazard-related information from the route                      | Algorithm to update the routes once the path is deemed safe |

Table 2.4: Event-Response Table for SafeRouteXplorer

## 3. System Overview

### 3.1 System Overview

The "Real-Time Safe Path Finder" system is designed to increase travel safety by distributing real-time hazard notifications and route safety information. It uses data collected from social media platforms such as Twitter and Instagram, and news channels to inspect hazards; topics of potential blockage or natural disaster- in road. Using NLP, the system filters and processes the relevant posts to create an Safety index for various routes. Those services can be approached via web and mobile applications like alert, map interactive, and customized safety preferences.

### 3.2 Architectural Design

#### 3.2.1 Box and Line Diagram

The initial system architecture can be visualized using a simple box and line diagram. This diagram shows the different modules of the system and how they interact with each other.

The modules include:

- **User Interface:** Allows users to access the system in order to view routes and also receive alerts as well as safety preference settings.
- **Data Collection:** Decide the external APIs like social media, news etc. in order to collect the information in real time.
- **NLP Processing:** Filters relevant posts and data to extract hazard-related information.
- **Safety Index Calculation:** Dynamically calculates the safety index for various routes based on the processed data.
- **Notification System:** Sends real-time alerts to users about hazards along their routes.
- **Database:** Maintaining list of user profiles, route information, and hazard alerts.

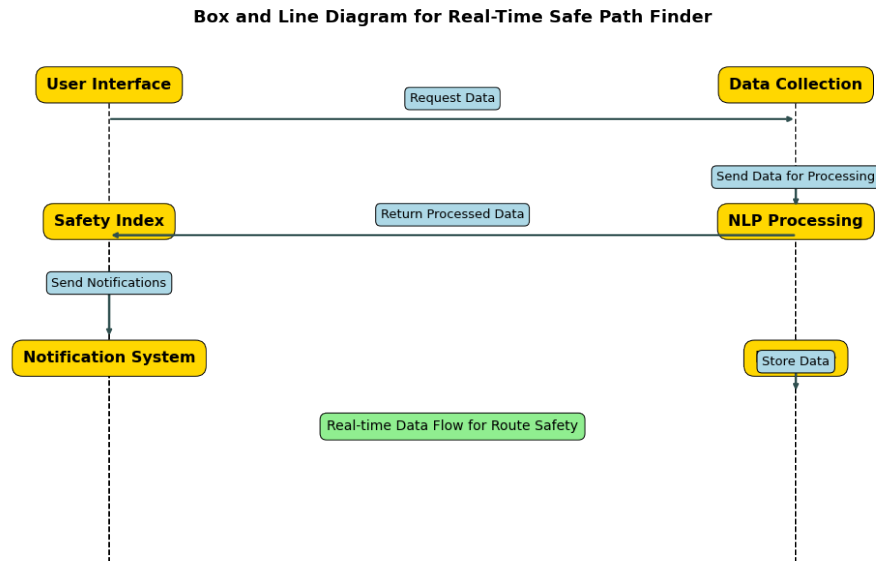


Figure 3.1: Box and Line Diagram for Real-Time Safe Path Finder

### 3.2.2 High-Level Architecture (Client-Server Style)

The system uses a client-server approach with web and mobile applications on the client side. Interaction with the server-side (back-end services) happens via APIs. The server-side com- Each unit will be responsible for the computation, safety index values, and real-time alerts. The back- terminate services access and gather social media posts and news data from external APIs. and which can be processed and stored in a real-time database. The architecture can also be described As multi-layered, it distinguishes between the layers of presentation, logic, and data.

## 3.3 Design Models

The design models provide a detailed view of how the system operates at different levels, including object-oriented and procedural approaches. The following models are used for the "Real-Time Safe Path Finder" project:

### 3.3.1 Activity Diagram

An activity diagram shows the high-level flow of the system, including user interactions, data processing, and real-time hazard alerts.

### 3.3.2 Class Diagram

The class diagram represents the structure of the system by showing its classes and the relationships between them.

### 3.3.3 Class level Sequence Diagram

For a **Class-Level Sequence Diagram**, we focus on the interaction between different classes or objects in the system.

### 3.3.4 State Transition Diagram

A state transition diagram indicates how a system transitions from one state to the next based on either user action or data processing events. Key States:

- **Normal State:** The system is running with no critical alerts as of now.
- **Data Processing State:** System is collecting and processing data from social media.
- **Alert State:** Hazard detected, and system sends alerts to users.
- **Feedback State:** Users provide feedback on route safety.

## 3.4 Data Design

The system utilizes a real-time database to store, process, and retrieve key data entities in the system. The following are major data structures currently being utilized below:

- **User Profiles:** Maintains user-specific information, preferences, location, and their preferred custom alert settings.
- **Routes:** Stores route information that includes start and end points, safety index, and other hazard information.
- **Alerts:** Stores real-time hazard alerts along with time-stamped, location, and hazard type.
- **Social Media Posts:** Stores social media posts that contain relevant hazard data.
- **Feedback:** Store feedback from users on route safety and hazard accuracy.

All data is processed using Firebase NoSQL database to ensure real-time synchronization and availability for users.

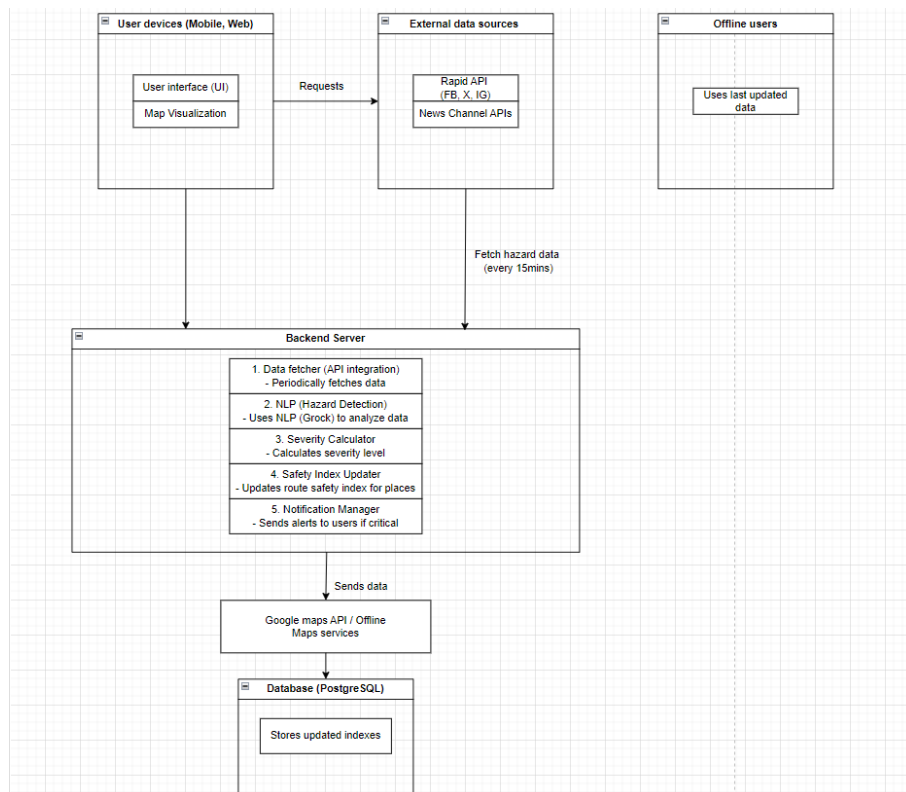


Figure 3.2: High-Level Architecture for Real-Time Safe Path Finder

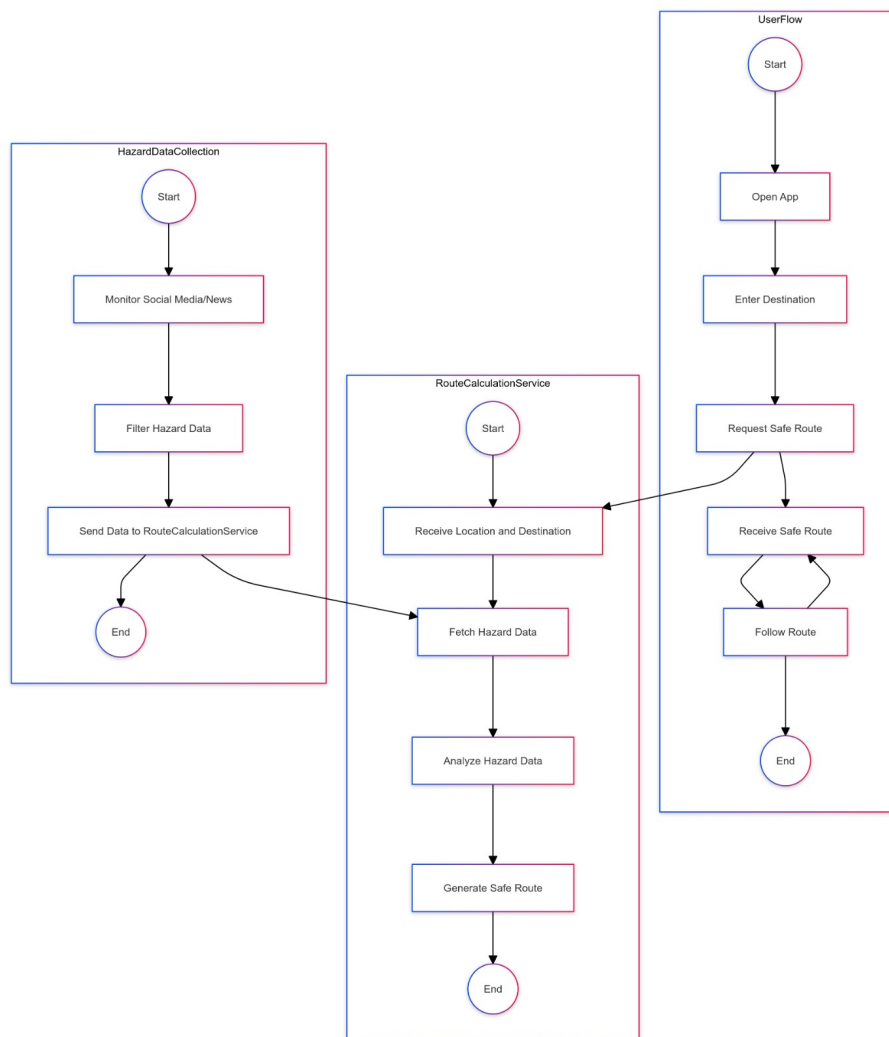


Figure 3.3: Activity Diagram for Real-Time Safe Path Finder

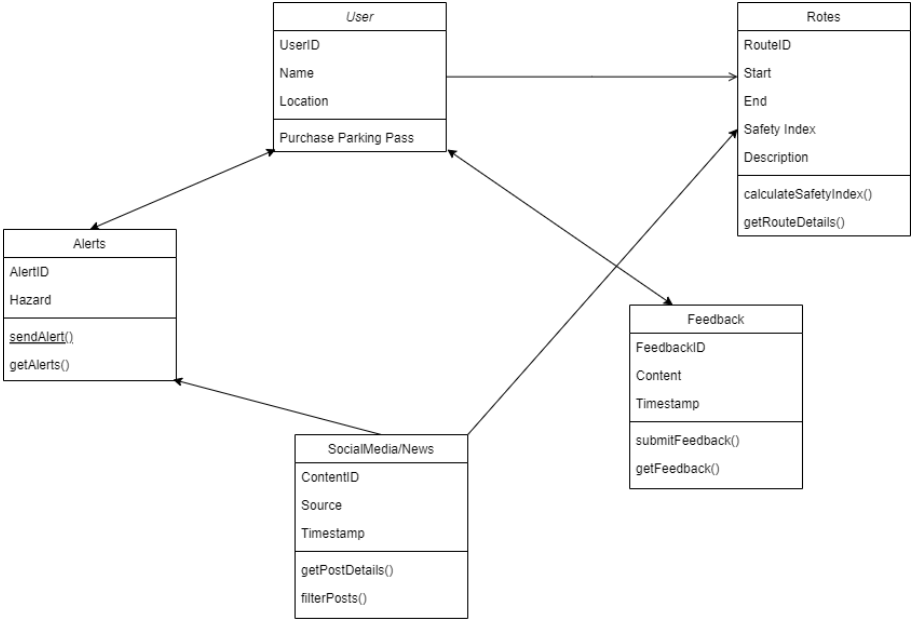


Figure 3.4: Class Diagram for Real-Time Safe Path Finder

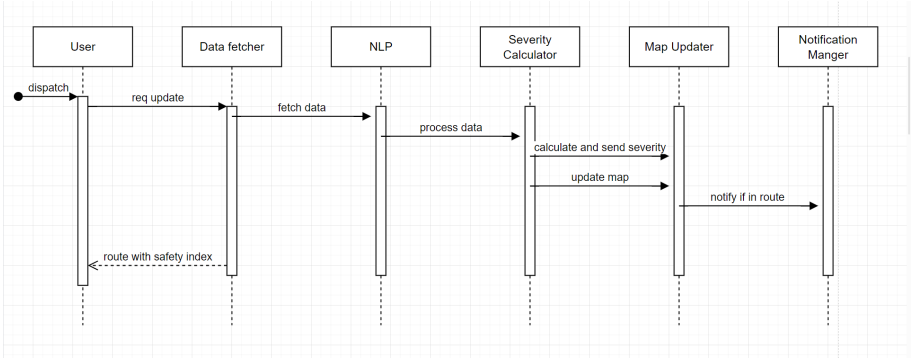


Figure 3.5: Activity Diagram for Real-Time Safe Path Finder

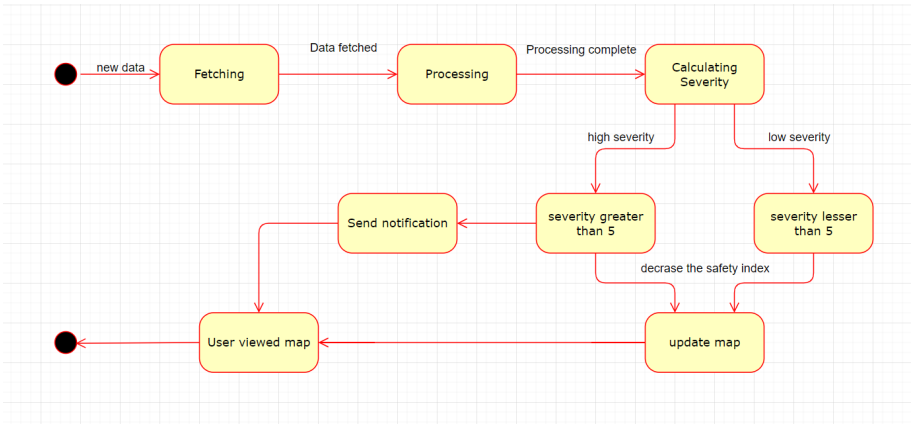


Figure 3.6: State Transition Diagram for Real-Time Safe Path Finder